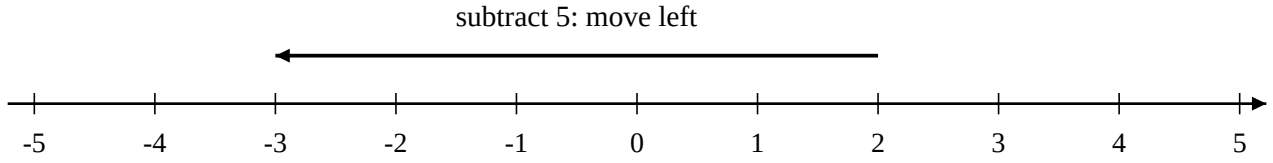
Numbers, Decimals, and Fractions

1. Negative numbers: move on the number line



2 - 5 = -3 Subtracting a negative means add: **4 - (-3) = 4 + 3 = 7**.

Multiply/divide: same signs → positive; different signs → negative.
(-3) × (-2) = 6 (-12) ÷ 3 = -4

Practice: 1. 9 + (-14) 2. 2 - (-5) 3. -18 ÷ 6

2. Decimals: line up places; estimate the answer

Subtract

```
  6.00
- 2.75
------
  3.25
```

Multiply

1.2 × 0.4
12 × 4 = 48
Two decimal places:
0.48

Divide

7.2 ÷ 0.6
Shift both decimals:
72 ÷ 6 = **12**

Practice: 4. 5.4 - 1.85 5. 0.3 × 8 6. 4.5 ÷ 0.5

3. Fractions are division

$\frac{3}{4}$ = 3 ÷ 4 = 0.75



3 of 4 equal parts

Add: use the same denominator. 1/2 + 1/4 = 2/4 + 1/4 = 3/4

Multiply tops and bottoms. (2/3) × (3/4) = 6/12 = 1/2

Divide: flip the second fraction. (3/4) ÷ (1/2) = (3/4) × (2/1) = 1.5

A fraction bar groups the entire top: **(9 - 3)/2 = 6/2 = 3**.

Practice: 7. 3/5 as a decimal 8. (9 - 3)/2

9. (1/2) × (2/3) 10. (2/3) ÷ (1/3)

Check

1: -5; 2: 7; 3: -3; 4: 3.55; 5: 2.4; 6: 9; 7: 0.6; 8: 3; 9: 1/3; 10: 2.

Equations, Order, and Units

1. Solve an equation: do the same thing to both sides

$$x + 7 = 19 \quad \downarrow \text{ subtract 7 from both sides} \quad x = 12$$

Check: 12 + 7 = 19.

$$4x = 18 \quad \downarrow \text{ divide both sides by 4} \quad x = 4.5$$

Check: 4 × 4.5 = 18.

Undo addition with subtraction. Undo multiplication with division.
For x/5 = 3, multiply both sides by 5: **x = 15**.

Practice: 11. x + 8 = 15 12. 6x = 21 13. x/4 = 2.5

2. Substitute, then follow the order

Parentheses → exponents → multiply/divide → add/subtract.
Work left to right within each multiplication/division or addition/subtraction step.

$$a = \frac{2 - 8}{3} \longrightarrow \frac{-6}{3} \longrightarrow \mathbf{-2}$$

Do not enter 2 - 8 ÷ 3; that loses the parentheses.
In acceleration, divide the change in velocity by the elapsed time.

Practice: 14. (4 - 10)/2 15. 3 + 0.5 × 8

3. Convert units before calculating

minutes $\xrightarrow{\times 60}$ **seconds**

1.5 min × 60 = 90 s

grams $\xrightarrow{\div 1{,}000}$ **kilograms**

750 g ÷ 1,000 = 0.75 kg

1 km = 1,000 m. For m/s, use meters and seconds. For F = ma, use kilograms.
m/s² means “meters per second, each second.” It describes acceleration.

Practice: 16. 2.5 min in seconds 17. 400 g in kilograms

18. 0.8 km in meters

Finish: round only at the end. Include the unit and direction when needed.

Check

11: 7; 12: 3.5; 13: 10; 14: -3; 15: 7; 16: 150 s; 17: 0.4 kg; 18: 800 m.